

Sampling and Confidence Intervals

1 Sampling

Let us assume a large population of size N possesses an underlying, unknown true mean μ and an unknown true variance σ^2 .

First, we begin by drawing a single random sample of size n from this population. We observe individual data points and compute their arithmetic average, which we denote as the first sample mean, $\bar{X}_1$.

Then, we again draw a completely new random sample of the exact same size n . We calculate the average of this second group, obtaining a second sample mean, $\bar{X}_2$.

Third, we repeat this sampling loop an infinite number of times. This process yields an entire collection of thousands of distinct sample means: $\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots, \bar{X}_\infty$.

Fourth, instead of analyzing individual data points, we treat this massive spreadsheet of sample means as our new dataset. When we plot these sample means as a probability density function, we create the sampling distribution of the sample mean, denoted as $\bar{X}$.

The variance of these sample means measures how much different sample averages fluctuate from one sample to the next. When accounting for any sample size n and finite population size N , the true variance of the sample means is given by the formula:

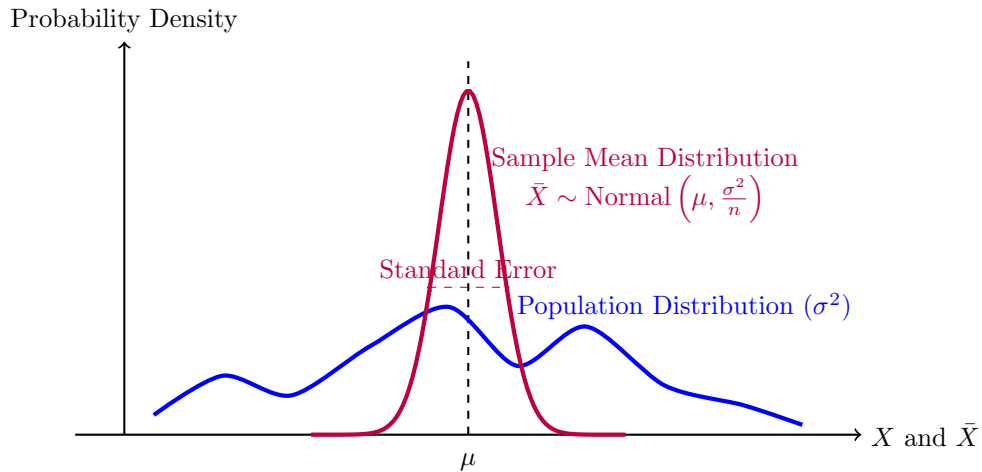
$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n+1}{N+1} \right)$$

This formula demonstrates that as the sample size n grows, the denominator increases, forcing the overall variance of the sample means to shrink. This reflects the reality that larger samples resist extreme variations and cluster more tightly around the true mean. If the number N is extremely large, we get the formula for standard error.

2 The Standard Error of the Sampled Means

The Standard Error represents nothing more than the standard deviation of the sampling distribution of the sample means. While the standard deviation σ measures the spread of individual observations within the population, the Standard Error measures the spread of various sample averages.

The diagram below visually contrasts the wide, irregular distribution of the underlying population with the narrow, smooth, symmetric normal distribution of the sampled means centered at μ .



As shown, even if the population distribution exhibits irregular peaks and wide dispersion, the distribution of the sample means converts into a clean, bell-shaped normal curve due to the Central Limit Theorem.

3 Shifting and Scaling the Sampled Means

To make the distribution of the sample mean mathematically workable across any arbitrary unit of measurement, we perform a two-step transformation known as standardization. This process alters the center and the width of our pink curve to create a universal reference standard.

The first step is shifting the distribution. We take every single calculated sample mean $\bar{X}$ from our collection and subtract the true population mean μ from it:

$$\text{Shifted Value} = \bar{X} - \mu$$

Because the original distribution of $\bar{X}$ was centered perfectly at μ , subtracting μ from every data point physically slides the entire curve across the horizontal axis until its center peak rests exactly on top of 0. Therefore, the mean of this shifted distribution becomes zero.

The second step is scaling the distribution. We take the shifted values and divide them by their own standard deviation, which is the Standard Error ($\sigma/\sqrt{n}$):

$$\text{Standardized Value (Z)} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

By dividing the shifted distance by the natural unit of spread for the sample means, we compress or stretch the width of the distribution. Dividing any distribution by its own standard deviation forces its new standard deviation to equal exactly 1. Because variance is the square of the standard deviation, the new variance also becomes exactly 1.

Consequently, this shifting and scaling transformation alters our variable into a Standard Normal Distribution, expressed as:

$$Z \sim \text{Normal}(0, 1)$$

The horizontal axis no longer reads in original raw metrics; instead, it measures values in terms of standard deviations away from the center.

4 The True Meaning of a Confidence Interval

In practical application, an investigator cannot collect thousands of samples. We collect exactly one single sample, calculating one single sample mean $\bar{X}$ and one single sample standard deviation s . We use this solitary sample to construct a single interval boundary using the formula:

$$\bar{X} \pm \left(Z_{\text{critical}} \times \frac{s}{\sqrt{n}} \right)$$

To construct a confidence interval from our single sample, we must determine exactly how wide our interval needs to be to match our desired level of certainty. This width is determined by finding the appropriate critical value, known as the Z -value, from the Standard Normal Distribution.

The mathematical process for defining this Z -value unfolds as follows:

First, we define our desired confidence level as p (expressed as a decimal, such as 0.95 for a 95% confidence interval). The remaining probability outside of our confidence interval is called the significance level, denoted by the Greek letter alpha (α). We calculate this alpha value using the relationship:

$$\alpha = 1 - p$$

For a 95% confidence interval, $\alpha = 1 - 0.95 = 0.05$. This represents the total 5% probability of pulling a unrepresentative "dud" sample mean that lands in the extreme outer tails of our distribution.

Second, because a standard normal curve is perfectly symmetric and our confidence interval is two-sided, the total risk of error (α) must be divided equally between the lower tail and the upper tail. Therefore, the area trapped in each individual tail is exactly half of alpha:

$$\text{Tail Area} = \frac{\alpha}{2}$$

For our 95% interval, each tail contains exactly $\frac{0.05}{2} = 0.025$ (or 2.5%) of the total probability.

Third, we locate the specific cutoff points on the horizontal axis of the Z -distribution that separate these extreme tail areas from the central p area. This critical value is denoted mathematically as:

$$Z_{\alpha/2}$$

This notation represents the exact number of standard deviations you must travel away from the center of a standard normal distribution to leave a probability area of exactly $\alpha/2$ in the outer tail. For a 95% confidence interval, $Z_{0.025} = 1.96$.

Fourth, to apply this universal standard deviation multiplier to our actual data, we scale it by multiplying it directly with the Standard Error (SE) of our sample means. This calculates our physical Margin of Error (E):

$$E = Z_{\alpha/2} \times \text{SE}$$

Substituting our formula for the estimated Standard Error ($s/\sqrt{n}$), the final boundary calculation for our real-world sample mean ($\bar{X}$) is expressed as:

$$\text{Confidence Interval} = \bar{X} \pm \left(Z_{\alpha/2} \times \frac{s}{\sqrt{n}} \right)$$

By multiplying the Standard Error with $Z_{\alpha/2}$, we successfully translate our theoretical probability requirements into the exact physical units of our data, creating a custom net engineered to capture the true population mean μ with a historical success rate of p . To understand what a 95% confidence interval truly means, we must relate our single real-world sample back to our theoretical multiple-sample framework using the property of mathematical symmetry.

Symmetry dictates that the distance from the fixed true mean μ to an observed sample mean $\bar{X}$ is identical to the distance from that sample mean $\bar{X}$ back to the true mean μ .

Looking at the theoretical distribution of sample means, we know that if we take repeated samples, 95% of all resulting sample means will land close to the center, falling within a specific threshold distance of the true mean μ . The remaining 5% of sample means will land far out in the extreme tails of the distribution.

When we translate this to the textbook definition of a confidence interval, it means that if we take repeated samples and construct a unique confidence interval for each individual sample, 95% of those constructed intervals will successfully capture the true, fixed value of the parameter μ .

This occurs because 95% of the time, our sample mean lands close to the true center, meaning an interval built around that sample mean will successfully stretch far enough to cover μ . Conversely, 5% of the time, our sample mean will be an unrepresentative outlier from the tails, and an interval built around it will fail to reach the true mean.

Therefore, when we calculate an interval from our single real-world sample, the interval itself is fixed and no longer possesses a shifting probability. We

state we are 95% confident because we trust the reliability of the overarching mathematical process, knowing that 95% of all intervals generated by this exact methodology successfully trap the hidden population parameter.